PMP-30 Circulation Pump - Quick Service Guide

1. About this guide

Synthetic reference text written for the AlixAI facility-maintenance reference application.
No real equipment is described.
The PMP-30 is a close-coupled circulation pump with a 30 mm mechanical seal.

2. Normal operating values

- Bearing vibration: below 4.5 mm/s RMS. Above 7 mm/s, stop the pump and inspect the alignment.
- Seal leakage: a few drops per minute are normal while the seal beds in. A steady drip means the seal must be replaced.
- Motor current: within 10 percent of the nameplate value.

3. Replacing the mechanical seal

Part SEL-30 Mechanical pump seal 30 mm. Isolate the pump electrically and close both isolation valves. Drain the casing, remove the motor and impeller, and slide the old seal off the shaft. Clean the shaft, fit the new seal with a drop of water as lubricant (never oil) and reassemble.
Do not touch the seal faces with bare fingers. Refill, vent the casing and check for leaks at full pressure.

4. Parts

- SEL-30 Mechanical pump seal 30 mm, quantity per change: 1.
- CAP-45 Run capacitor 45 uF.